WORKED SOLUTION

STEP III 1999, Question 8

Pure • Piecewise Differential Equations

STEP III 1999, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q8

The key idea

On each interval the equation is simple harmonic motion with a constant frequency, so the general solution there is a combination of $\cos nx$ and $\sin nx$. The only work is matching: the continuity conditions hand each interval its starting values of y and y' from the end of the previous one. The intervals all have length 2π, so the starting values turn out to be the same every time, and a pattern appears that we can prove by induction.

Solution

Part (i)

For $0 < x < 2\pi$ we have $n = 1$, so $y'' + y = 0$, with general solution

$$y = A\cos x + B\sin x.$$

Since y and y' are continuous at $x = 0$, the conditions $y(0) = 0$ and $y'(0) = 1$ apply to this solution: $y(0) = A = 0$ and $y'(0) = B = 1$. Hence

$$y = \sin x \qquad (0 \le x \le 2\pi).$$

At the right-hand end, $y(2\pi) = \sin 2\pi = 0$ and $y'(2\pi) = \cos 2\pi = 1$.

Part (ii)

For $2\pi < x < 4\pi$ we have $n = 2$, so $y'' + 4y = 0$ and

$$y = A\cos 2x + B\sin 2x.$$

By continuity of y and y' at $x = 2\pi$, this solution must have $y = 0$ and $y' = 1$ there, the values found at the end of part (i). Since $\cos 4\pi = 1$ and $\sin 4\pi = 0$,

$$y(2\pi) = A = 0, \qquad y'(2\pi) = 2B\cos 4\pi = 2B = 1,$$

so $B = \frac{1}{2}$ and $y = \frac{1}{2}\sin 2x$ for $2\pi \le x \le 4\pi$, as required.

Notice that again $y(4\pi) = \frac{1}{2}\sin 8\pi = 0$ and $y'(4\pi) = \cos 8\pi = 1$: the next interval starts from exactly the same values. This suggests the following claim.

Claim. For every positive integer n,

$$y = \frac{1}{n}\sin nx \qquad \text{for } 2(n-1)\pi \le x \le 2n\pi.$$

Proof by induction. The case $n = 1$ is part (i). Suppose the claim holds for $n = k$. At $x = 2k\pi$ the previous piece gives

$$y = \frac{1}{k}\sin(2k^2\pi) = 0, \qquad y' = \cos(2k^2\pi) = 1,$$

because $2k^2\pi$ is a whole multiple of 2π. On $2k\pi < x < 2(k+1)\pi$ the equation is $y'' + (k+1)^2 y = 0$, so

$$y = A\cos(k+1)x + B\sin(k+1)x.$$

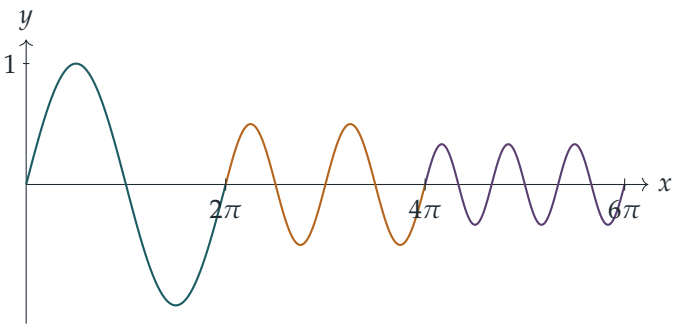
Ask yourself What are $\cos(k+1)x$ and $\sin(k+1)x$ at the left-hand end $x = 2k\pi$?

There $(k+1)x = 2k(k+1)\pi$, again a whole multiple of 2π, so the cosine is 1 and the sine is 0. Continuity of y and y' at $x = 2k\pi$ therefore gives

$$A = 0, \qquad (k+1)B = 1,$$

so $y = \dfrac{1}{k+1}\sin(k+1)x$ on $2k\pi \le x \le 2(k+1)\pi$. This is the claim for $n = k+1$, and by induction it holds for all $n \ge 1$. □

So $y(x) = \dfrac{1}{n}\sin nx$ whenever $2(n-1)\pi \le x \le 2n\pi$. (At each junction $x = 2n\pi$ both neighbouring formulas give $y = 0$, so there is no ambiguity.)



Part (iii)

Using part (ii), on the nth interval

$$\int_{2(n-1)\pi}^{2n\pi} y^2\,\mathrm{d}x = \frac{1}{n^2}\int_{2(n-1)\pi}^{2n\pi} \sin^2 nx\,\mathrm{d}x = \frac{1}{2n^2}\int_{2(n-1)\pi}^{2n\pi} (1-\cos 2nx)\,\mathrm{d}x = \frac{1}{2n^2}\cdot 2\pi = \frac{\pi}{n^2},$$

since $\sin 2nx$ vanishes at both limits (each limit is a multiple of π and $2n$ is an integer), so the cosine term integrates to zero.

Adding the first N intervals,

$$\int_0^{2N\pi} y^2 \,\mathrm{d}x = \pi \sum_{n=1}^{N} \frac{1}{n^2}.$$

The infinite integral is the limit of $\int_0^X y^2 \,\mathrm{d}x$ as $X \to \infty$. Since $y^2 \geq 0$, this is an increasing function of X; and for $2(N-1)\pi \leq X \leq 2N\pi$ it lies between the values at $2(N-1)\pi$ and $2N\pi$. Both of those tend to $\pi \sum_{n=1}^{\infty} \frac{1}{n^2}$ as $N \to \infty$, because the series converges. Hence

$$\int_0^{\infty} y^2 \,\mathrm{d}x = \pi \sum_{n=1}^{\infty} \frac{1}{n^2},$$

as required.

Checks

- Each piece $\frac{1}{n}\sin nx$ satisfies $y'' = -n^2 y$, starts at $y = 0$, $y' = 1$ and ends at $y = 0$, $y' = 1$, so the pieces join smoothly in value and slope.
- Stepping the differential equation numerically (fourth-order Runge–Kutta) through the first six intervals reproduces $\frac{1}{n}\sin nx$ to within 10^{-10}, and the integral of y^2 up to 12π comes out as 4.685336, matching $\pi(1 + \frac{1}{4} + \cdots + \frac{1}{36})$.
- Using Euler's result $\sum 1/n^2 = \pi^2/6$, the integral equals $\pi^3/6 \approx 5.168$.

Common slips

- Applying the initial conditions at $x = 0$ on every interval, rather than at its left-hand end $x = 2(n-1)\pi$. It happens to give the right answer here only because the starting point is a multiple of 2π, and that fact must be stated.
- Forgetting that $y' = nB\cos nx$: the factor n is exactly what produces the amplitude $\frac{1}{n}$.
- Writing the integral as an infinite sum without saying why the limit exists; the positivity of y^2 makes this easy.

Beyond STEP

This is a simple example of a differential equation with piecewise-constant coefficients, the kind that models a spring whose stiffness is switched at fixed times; at each switch the solution is still continuous and differentiable even though y'' jumps. The final answer, $\pi^3/6$, ties a differential equation to the Basel sum $\sum 1/n^2 = \pi^2/6$, which is often proved first in undergraduate courses on Fourier series using Parseval's theorem.

Boyce and DiPrima, *Elementary Differential Equations*, covers both second-order linear equations and Fourier series.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.